

MAINS MASTER PROGRAM (MMP) 2024

DISASTER MANAGEMENT-2

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2. LANDSLIDES

- Example Questions

- » While the **Himalayan region was always vulnerable to landslides**, the recent years have seen the peninsular hills also becoming increasingly prone to this natural disaster. Give reasons. What are the NDMA guidelines for management of landslides? [15 marks, 250 words]

- Introduction

- » Landslide is defined as the **movement of a mass of rock, debris or earth down a slope**. This is a type of mass wasting, which denotes any down-slope movement of soil rocks under the direct influence of gravity. The term "landslide" encompasses five modes of slope movement: Falls, topples, slides, spreads, and flows.
- » There are **two landslide hotspots** that exist in India - along the **southern edge of the Himalayan arc** and the **Western Ghats region**.

- Key causes of landslides:

- » **Himalayas** are prone to landslides because of several morphological and geological factors like:
 - Tectonic movement
 - Glacial movements
 - Freeze and thaw effect
 - Unstable rock structure
 - Steep slopes etc.
 - Most regions with more than 20 degrees are prone to landslides.
 - Types of rocks, weaknesses, zone of rupture etc.
- » But in recent years, we have also seen the stable Peninsular Hills like in Western Ghats becoming vulnerable because of the human made factors.
- » **Human Induced Causes**
 - Climate Change: Extreme Rainfall Events trigger slope failure where lateral spread and soil piping have occurred.
 - For instance, unusually high rainfall in Kerala since Aug 2018 has destabilized the already vulnerable hill slopes in the high ranges and has caused many landslides.
 - Illegal Mining, deforestation etc have made the **surface weak and vulnerable to landslides**
 - For e.g. the railways have blamed two successive landslides along their project site in Manipur on the traditional practice of Jhum or shifting cultivation.
 - Unscientific Farming and Construction Activities
 - UNDP's assessment after the 2018 flood says that changes in land cover, blocking of natural drains, and poor agricultural practices such as monocropping have all exacerbated the risk of landslides in Kerala.
 - Illegal landgrab using fake deeds have contributed to unscientific land use and thus landslides.
 - Inadequate Early Warning Systems makes the impact of the disaster worse.

- Government Efforts and Way Forward” NDMA guidelines for management of Landslides

- » Hazard mapping and vulnerability analysis:

- **Steps Taken:** A national landslide susceptibility map has been created by Geological Survey of India (GSI). This should be integrated with infrastructure development and planning in hilly areas.
- » **Risk Reduction Efforts:** Afforestation; banning mining in sensitive areas, sustainable forms of agriculture etc.
- » **Establishing Early Warning System:**
 - **Steps Taken:** Union Ministry of Earth Science has also initiated steps to establish a network of landslip monitoring stations in the highlands. The units which will be based on acoustic emission technology will also have an early warning mechanism to alert the local community
- » **Awareness Generation and preparedness** among various stakeholders through setting up of institutional mechanisms.
- » **Capacity Building to deal with Landslides**
 - Enhancing education and training of professionals involved in landslides management.
 - Capacity development of organizations working in the field of landslides
- » **Create an Autonomous National Centre** for landslide studies research and management.

3. TSUNAMI

- **Example Questions**
 - On December 2004, tsunami brought havoc in 14 countries including India. Discuss the factors responsible for the occurrence of Tsunami and its effects on life and economy. In the light of guidelines of NDMA (2010) describe the mechanisms for preparedness to reduce the risk during such events. [15 marks, 250 words][CSM 2017]
 - Discuss the key steps taken by India towards Tsunami Disaster Preparedness since the 2004 Tsunami? [10 marks, 150 words]
- **Introduction**
 - The term Tsunami has been derived from the Japanese term '**Tsu**' meaning harbour, and '**nami**' meaning waves. Thus, tsunami means 'harbour waves'.
 - Tsunami consists of a series of waves which rise as high as 10 meters or more. They move inland, sometimes several 100 kms and cause untold disasters. These waves can move at very high speeds sometimes exceeding even 50 km/h on the coastal plains.
- **26th Dec 2004 Tsunami/Great Sumatran Andaman Earthquake /Asian Tsunami /Boxing Day Tsunami**
 - » An undersea earthquake occurred on 26th Dec, 2004, with an epicenter off the west coast of Sumatra, Indonesia. The magnitude of the earthquake was between 9.1 to 9.3 on the Richter scale, making it the second largest earthquake ever recorded on seismograph.
 - » It triggered the devastating Tsunami along the coasts of most of the countries bordering the Indian Ocean and killed 2,25,000 people.
- **Causes of Tsunami**
 1. **Earthquake** of more than 6.5 on the Richter Scale, with a vertical disruption of water column due to vertical tectonic displacement of the sea bottom along a zone of fracture in the earth's crust is the most important cause of Tsunami.

2. **Volcanic Eruption**, and **submarine landslides** are other reasons which may displace water and cause it to inundate coastal region.
3. **Nuclear Explosion, fall of large celestial bodies** (like asteroid, meteorites, comets etc can also cause Tsunami)

- **Tsunami Detection**

- i. **Different Ways to detect Tsunamis:** A natural disaster like Tsunami can't be stopped, but early warnings can help in saving people and containing the damage it can cause to a certain extent.

- **Coastal tidal gauges**
- **Satellite technologies** have now made it possible to detect tsunami genic earthquakes almost immediately.
- **Deep Ocean Assessment and Reporting of Tsunamis (DART) Project** was developed by the U.S. National Tsunami Hazard Mitigation Program. It strives an early Tsunami detection and real time reporting capability.

- ii. **State Preparedness Measure Before Tsunami**

- Hazard Mapping: Identify Tsunami prone areas
- Risk Reduction steps: Increasing afforestation (Mangroves etc) in coastal areas; Reducing the population in coastal areas; creating infrastructure in coastal region which is flood proof and Tsunami proof etc.
- Early Warning System
- Awareness Generation: Educate people in these regions about Tsunamis - inform them about the evacuation routes in case a Tsunami hits the coast.
 - Mock Evacuation drills should be conducted periodically so that people are aware of it
- Capacity Building to deal with Tsunami disaster
- Increased focus on R&D related to Tsunami and Tsunami response
- International Cooperation for faster information sharing.

- iii. **Preparedness by citizens**

- Live at safe distance from coastline; in elevated houses; well managed drainage system; easy access to information system like TV radio etc; following SOP during Tsunami.

- iv. **Tsunami Response**

- **People on Land:** After warning sound move to safer places with cattles (top floor of the multi-storied buildings); stay away from rivers that flow into the oceans; Listen to radio and TV for updates regularly.
- **People on Sea:** Don't return to coast; if time is available move to deeper waters;
- **Government:** Search and rescue; basic services like water, first aid etc.

- **Steps taken by India Since 2004 Tsunami:** The 2004 Tsunami highlighted the clear lack of preparedness about Tsunamis all over the world. None of the affected countries, including India had any system for early warning of such Tsunamis, nor was there any plan for emergency response. Learning a lesson from 2004 Tsunami, GoI have taken a number of steps:

1. **Early Warning System:**

- GoI has established the state-of-art India Tsunami Early Warning Centre (ITEWC) (operational since 2007) at the Indian National Centre for Ocean Information Services (INCOIS) as an autonomous body under Ministry of Earth Sciences.
 - INCOIS has extended **GIS-based 3D protocol on Tsunami warning to all vulnerable areas** in the country with new methodologies and improved warning procedures.
2. **Strengthening Tsunami Research** has been a key focus of GoI since 2004.
 3. **National Disaster Management Guidelines** for management of Tsunami, 2010
 - The guidelines include a wide range of approach for strengthening the Tsunami Preparedness and mitigation. It focuses on:
 - Tsunami Warning System
 - Capacity Building
 - Education
 - Building codes and Safety Standards
 - Land Use planning and other engineering solution
 - These guidelines are an important step towards comprehensive Tsunami Risk Management in India.
 4. **Regular Mock Drills** are conducted by NDMA, INCOIS and Ministry of Home Affairs
 - These exercises help familiarize participants with their responsibilities, actions required and further help them evaluate the Standard Operating Procedure (SOP) for Tsunami warnings.
 5. **Awareness generation programs** in coastal area is also conducted regularly since 2004.
- **Steps that we further needs to take**
- » **Improve EWS: Advanced technology** such as Artificial Intelligence and Machine learning should be incorporated to improve the Early Warning System in the country.
 - » **Increased International Collaboration in real time monitoring will** help support each other during emergencies.
 - BIMSTEC can play a very important role in promoting this collaboration in Bay of Bengal region.
 - » **Regular training and Capacity building through workshops, drills etc.** can help us avoid the 2004 scenario.
 - » **Land use planning** is one area which has been mostly ignored
 - There is a requirement of more vegetation cover in coastal region, but various studies have shown that mangrove cover has gone down over the years
 - » **Removing the limitations of India's disaster management institutional framework**

4. EARTHQUAKE

- Introduction

- » India lies at the northwestern end of the Indo-Australian Plate, which encompasses India, Australia, a major portion of the Indian Ocean and other smaller countries.
- » The major reason for the high frequency and intensity of earthquakes is that Indian plate is driving into Asia at a rate of approximately 47 mm/year.
- » According to NDMA, about 59% of India's land could **face moderate to severe earthquakes**.

- Different zones

- Bureau of Indian Standards, based on the past seismic history, grouped the country into **four seismic zones**, viz. Zone-2, Zone-3, Zone-4 and Zone-5. Of these, Zone 5, is the most seismically active region, while zone 2 is the least.

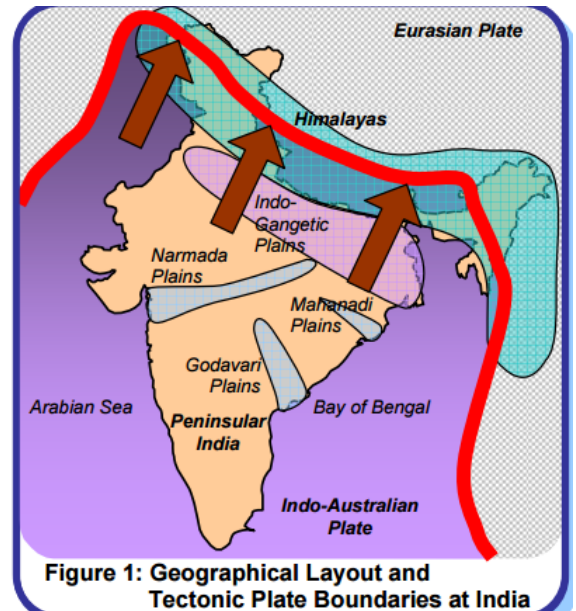
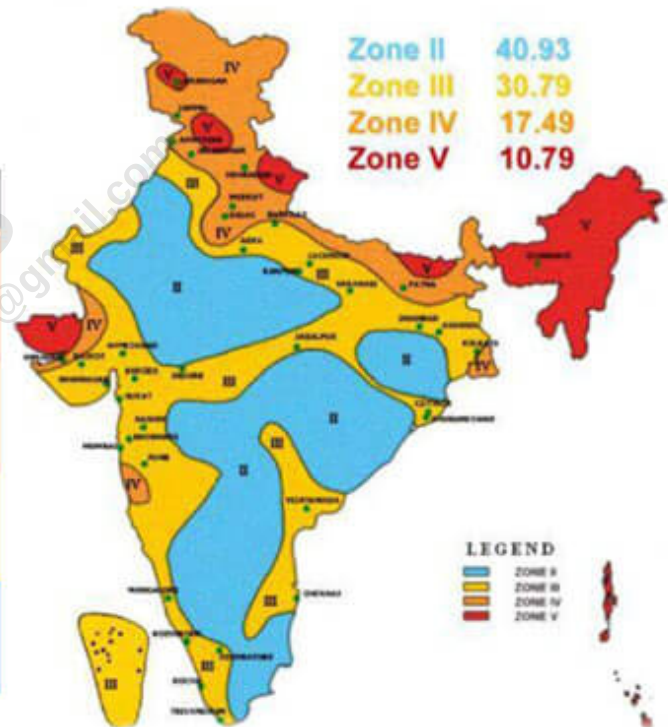


Figure 1: Geographical Layout and Tectonic Plate Boundaries at India

Seismic Zone, Map of India 2002

About 59 percent of the land area of India is liable to seismic hazard damage

Zone	Intensity
Zone V	Very High Risk Zone Area liable to shaking intensity IX (and above)
Zone IV	High Risk Zone Intensity VIII
Zone III	Moderate Risk Zone intensity VII
Zone II	Low Risk Zone VI (and lower)



- Causes of Earthquakes

- » The important causes of earthquakes can be divided into two categories: Natural and Manmade
 - i. **Natural Causes**
 1. **Tectonic Movements**
 2. **Volcanic Activities**
 3. **Adjustment in the inner rocks beds (Plutonic Earthquakes)**
 - Adjustment between Sima and Sial in the interior of the earth's crust.

4. Pressure of Gases in the interior

- The expansion and contraction of gases in the interior of the earth sometimes cause a sudden shake on the earth's surface

5. Other causes

- Landslides and avalanches
- Denudation of landmasses and deposition of materials

ii. Man-made causes

1. **Dams:** Impounding of large quantity of water behind the dams disturb the crustal balance and can cause earthquakes

2. **Nuclear Bombs**

- The shockwaves through the rocks set up by underground testing of Atom bombs or Hydrogen bombs may be severe enough to cause an earthquake.

iii. Causes which make India very vulnerable

- Fragile built environment (Similar sized earthquake in Japan or USA will cause much smaller damage)

- Consequences of Earth Quakes

- » Human and livestock loss
- » Damage to Property
- » Tsunamis
- » Change in river course - floods
- » Fountain of muds
- » Landslides

- NDMA Guidelines for Management of Earthquake: Six Pillars of Earthquake Management in India

- a. Ensure the incorporation of earthquake resistant design features for the construction of new structures.
- b. Facilitate selective strengthening and seismic retrofitting of existing priority and lifeline structures in earthquake-prone areas.
- c. Improve the compliance regime through appropriate regulation and enforcement.
- d. Improve the awareness and preparedness of all stakeholders.
- e. Introduce appropriate capacity development interventions for effective earthquake management (including education, training, R&D, and documentation).
- f. Strengthen the emergency response capability in earthquake-prone areas

- Earthquake hazard Reduction and Mitigations

- Earthquake is a natural phenomenon which is **difficult to prevent and therefore we can only prepare for the earthquake and post-earthquake response**. The foreknowledge of potential danger areas can help mitigate the impact of a disaster.

- Steps that needs to be taken

» Short Term Steps:

1. **Preparing Vulnerability Map** of the country
2. **Infrastructure improvement for Risk Reduction:**

- **Implementing Building Codes:** Municipalities need to ensure that BIS codes and guidelines are properly being followed in new buildings.
 - The lifeline buildings which have been built in the past and are not earthquake resistant needed to be upgraded by retrofitting techniques
 - **Improving building code** through a new architecture regime and this be made mandatory for all builders and developers.
3. **Strengthening Early Warning Infrastructure** -> for providing those crucial extra seconds of a minute in disaster scenario.
 4. **Educating people** through dissemination of information about the ways and means of minimizing the adverse impacts of earthquakes. Dissemination of techniques such as 'Drop, Cover and Hold' is very important reducing the loss of life during earthquakes.
 - **Community preparedness** is very important for dealing with earthquake. Most of the time it is the community which plays the first hand role in disaster situation and therefore they have to be given proper training regarding search and rescue as well.
 - Community should have a disaster emergency kit ready for disaster situation
 - Community also needs to participate in planning, implementation and monitoring process of any method being used
 5. **Strengthening Institutional Framework** -> Shortage of manpower in SDRF and NDRF should be rectified. Local governments, NGOs etc also need to be prepared for post disaster response especially in vulnerable areas.
- » **Long Term steps:**
 1. **International Collaboration**
 - We should collaborate with other countries in development of earthquake resistant infrastructure. Our collaboration with countries like Japan will be very crucial in enhancing our preparedness for earthquakes.
 2. **Decongesting of Cities** to reduce risk and vulnerabilities
 3. **Research and Development** to find new technologies for improved warning and reduce losses.
 - **Conclusion**
 - » Earthquake may not yet be predictable and certainly not preventable, still if effective and timely steps are taken, the adverse impact of Earthquakes can be considerably blunted.

5. HEATWAVES

- **Why in news?**
 - » In July 2024, Kashmir grappled with a rare heatwave triggered by day temperature that runs six degree Celsius above normal temperature
- **Example Questions**
 - » What are heat waves? Suggest a strategy to reduce India's vulnerability to heatwaves. [15 marks, 250 words]
- **Definition**
 - » A heat wave is a **period of abnormally high temperatures**, more than the normal maximum temperature that occurs during the summer season usually in the north-western parts of India. In India, heat waves typically occur between March and June, and in some rare cases extend till July.

- » **Indian Meteorological Department (IMD)** has given following criteria for heat waves.
 - **Maximum Temperature of at least 40 degree Celsius for Plains, 37 degree Celsius for coastal regions and atleast 30 degree Celsius for hilly regions.**
- » Following conditions are used declare heat waves:
 - a. **Based on Departure from Normal**
 - **Heat Wave:** Departure from normal is 4.5 degree to 6.4 degree
 - **Severe Heat Wave:** Departure from normal is > 6.4 degree
 - b. **Based on Actual Maximum Temperature (for plains only)**
 - **Heat Wave:** When actual maximum temperature $\geq$ 45 degree Celsius
 - **Severe Heat Wave:** When actual maximum temperature $\geq$ 47 degree Celsius.
- **Note:** Heat wave has not been notified as a disaster by the Government of India yet and hence is not eligible for relief under National/State Disaster Response Fund norms.
- **Increasing cases of Heat Waves in India:** According to Lancet Report, India faced 60 million heatwave exposure events in 2016, a rise from 40 million exposures in 2012. Similarly, the average length of heat waves in India ranged from 3-4 days, which is more than double of global average of 0.8 - 1.8 days. The key factors responsible for this are:
 - » **Climate change -> higher temperatures:** According to a report by UNICEF "*The Coldest year of the Rest of Their Lives*" - nearly every child will face frequent heatwaves by 2050.
 - » **Sparser Pre Monsoon shower and Delayed Monsoon:** This whether pattern coupled with El-Nino effect, which often increases temperature in Asia, combine to create the record high temperatures.
 - » The **Loo (hot and dry winds) originating from Pakistan and Northwest India**, has also contributed to increasing temperature in India.
 - » **Urbanization and its problems like Urban Heat Island (UHI) Effect** exacerbates the problem of heat wave in many parts of our country.
 - » **Decreasing Tree Covers** -> concrete jungles, land heats up more.
 - » **Decreasing water bodies** like lakes, ponds etc.
- **Impact of Heatwaves**
 - » **Health Impacts:** Prolonged exposure to heatwaves can result into dehydration, cardiovascular diseases, respiratory diseases and even death. Children, elderlies and those with pre-existing morbidities are particularly vulnerable.
 - According the NDMA, more than 24,000 people have died in India due to heat waves between 1992-2015.
 - » **Economic Loss:** According to Lancet, the output of workforce in India declined by 7%, equivalent to 75 billion labor hours every year.
 - » **Worsening of air pollution problems:** -> increased electricity use -> more fuel burned.
 - » **Social Impacts:**
 - **Vulnerable sections** like daily wage earners, farm labourers, women etc are disproportionately affected by extreme heat, largely because of unequal power dynamics and unequal access to resources.
 - Heatwave is limiting the time children are engage with their peers and the world around them through play.

- **Steps Taken So Far**
 - » The **IMD** has regularly issued heat wave warnings in different parts of the country to make people aware of the worsening situation.
 - » The **NDMA** has suggested things like covering of head, cross-ventilating rooms and sleeping under a slightly wet sheets.
 - » **Heat Action Plans (HAPs)** have been prepared by governments at various level - state, district, city etc. HAPs aim to increase preparedness and lower the adverse impacts of extreme heat by outlining strategies and measures to prepare for, address, and recover from heatwaves.
- **NDMA's revised guidelines for prevention and management of Heat Waves in India (Oct 2019)**
 - **Aim/Objective**
 - The guideline aims to provide framework for developing Heat Action Plans for implementation, inter-agency coordination and impact evaluation of heat wave response activities in cities/towns.
 - **Developing a Heat-wave Plans**
 - Generating heat wave risk and vulnerability map and mapping hotspots for developing a strategic mitigation action plan.
 - The Scope of heatwave needs to be expanded to accommodate humid heat and warmer nights in addition to extreme dry heat. This requires development of heat index that accounts for multiple factors beyond temperature.
 - Identifying **Vulnerable Population** - elderlies, pregnant women, chronic disease patient, resident of a particular type of housing, certain type of occupations etc.
 - Identification and Evaluation of factors leading to disproportionate increase in temperature in the city
 - **Risk Reduction Measures:**
 - Reducing Temperature in the cities through vertical gardens, small parks with water fountains, increased afforestation etc.
 - Curb Future UHI manifestation by incorporating findings from the built environment assessment
 - Adhere to city building codes.
 - **Other Steps:** Reviewing the existing occupational health standards, labor laws, and sector regulation for worker's safety.
 - **Special focus on farm laborers** as the agricultural sector was more vulnerable compared to the industrial and service sectors because workers there were more likely to be exposed to heat.
 - Establish Early Warning System and Communication Systems
 - **Making communities more aware and resilient** to after effects of the heatwaves.
 - **Capacity Building** for better response
 - Preparedness at the local level for health eventualities.
 - Health care system capacity building
 - Developing inter-agency response plan and coordination in the field.
 - Collaboration with private and Non-Government and Civil Society

- Coordinate with Research institutions for better built environment.
 - Government budget should allocate funds for R&D in this field
- **International Cooperation:** Internationally, the **global community** should work towards achieving the climate change mitigation goals by working towards Paris Climate targets and making the NDCs more ambitious.
- **Other Steps that can be taken:**
 - » **Response:**
 - Ensuring quick advanced communication and guidelines during heatwave condition.
 - Drinking water supply should be increased along the roadside during heatwave conditions
 - Health facilities should respond with all the relevant facilities.

6. DROUGHT

- **Important Quotes:**
 - "Indians know that the Monsoon is the real finance minister of India" - Environmental Activists Sunita Narain
- **Intro:** Drought is a period of below average precipitation in a given region, resulting in prolonged shortages of its water supply, whether atmospheric, surface water or ground water. A drought can last for months or for years.
- **Types of droughts**
 - » **Meteorological drought** is brought about when there is a prolonged time with less than average precipitation. Meteorological drought usually precedes other kinds of drought.
 - » **Agricultural droughts** are droughts that affect crop production or ecology of the range. They are caused by **shortfall in water available to the crops**. It can be caused by extended period of low precipitation, poor water management, soil erosion or other such situations.
 - » **Hydrological drought** is brought about when water reserves available in sources such as aquifers, rivers, lakes and reservoirs fall below the statistical average. Hydrological drought tend to show up more slowly because it involves stored water that is used but not replenished. Like an agricultural drought this can be triggered by more than just a loss of rainfall.
 - » **Socioeconomic droughts** occur when water shortage starts to impact people's lives, individually and collectively.
- **Causes**
 - **Natural Factors**
 - **Precipitation deficiency**
 - **El Nino Southern Oscillation:** All the severe meteorological and hydrological droughts between 1870-2018 were found to be caused due to positive phase of ENSO (El Nino Southern Oscillation)
 - **Indian Ocean Dipole Effect (IOD):** IOD, also known as the Indian Nino is an irregular oscillation of sea-surface temperature, in which the **Western Indian Ocean becomes alternately warmer and then colder than the eastern part of the ocean**, affecting the Indian monsoon

- Lack of pre-Monsoon shower. For e.g. in 2019, India witnessed the second driest pre-monsoon season in 65 years.
 - **Dry Season**
 - **Land Degradation** - Desertification, erosion etc.
- **Anthropological factors**
 - **Poor water management**
 - Subsidies on equipment and electricity usage has encouraged over-exploitation of ground water.
 - Surfeit of dams have wreaked havoc on riverine system.
 - Poor rainwater conservation - Currently India captures only 8% of its rainfall - one of the lowest in the world.
 - Too much focus on **water consuming power generation** (like coal based power plants)
 - **Agricultural inefficiencies** - Agriculture consumes more than 90% of India's water use. 80% of this water is used for water guzzling crops like rice, wheat and sugarcane. Further, less penetration of technologies like drip irrigation and other forms of micro-irrigation also leads to inefficient water utilization.
 - **Improper and Unsustainable implementations of Watershed Development Programs**
 - **Water pollution**: - India ranks 120th among 122 countries in a global water quality index.
 - **Climate Change** - The global temperature is already higher by more than 1 degree Celsius from the pre-industrial era. This has also contributed in the spell of drought in India. For instance, drought continued in India post 2016 despite a change from El-Nino conditions due to climate change.
- **Impacts of Droughts in India:**
 - i. **Physical - Geographical-Environmental Impact:**
 - Meteorological drought adversely affects the recharge of soil moisture, surface runoffs and ground water. Rivers, lakes, ponds etc. tend to dry up.
 - **Exacerbates ground water extraction and depletion**
 - Increases water and soil pollution - for instance deeper borewell have higher chances of arsenic and fluoride contamination.
 - ii. **Economic Impact:** According to MoEF&CC - desertification, land degradation and drought cost India nearly 2.5% of GDP every year.
 - iii. **Impact on Agriculture** - Large percentage of agriculture rain dependent -> reduction in agri output
 - shortage of food and other agri-produce -> **inflation**
 - Reduced farmers income -> **increased farmer distress** -> increased farmer suicide
 - iv. **Other Economic and Social Impacts**
 - **Water Security:** Scarcity of drinking water -> **Health issues**
 - **Energy supply** may get impacted if the country increases its dependency on hydropower. (Note: India gets 17% of its electricity from Hydropower)
 - **Loss of livelihood -> unemployment -> Poverty -> Distress migration to cities, sale of property & livestock**
 - **Slowing down of secondary and tertiary activities** due to fall in agricultural production and decline in purchasing power.

- **Increasing inequality** -> Drought hampers weaker section of society including farmers, landless workers, weavers, artisans etc.
- **Social stress and tension**, disruption of social institutions and increase in social crime
 - Growth in superstition, increasing belief in supernatural powers etc.
- v. **Increased inter-state and International river water dispute** -
 - e.g. the Cauvery water dispute between Karnataka and Tamil Nadu exacerbated during less rainfall year.
 - E.g. the disputes between India and China for the water distribution of 10 major rivers originating in Tibet
- **Drought Relief Measures / Coping with droughts**
 - Management of drought has now been outlined in much elaborate manner in the drought manual issued by the ministry of Agriculture:
 - **Drought Monitoring**;
 - **Contingency Crop Planning**;
 - **Relief Employment**;
 - **Water Resource Management**
 - **Food Security**: It is provided through food for work programs etc.
 - **Relief through tax waivers and concessions**
 - **Cattle Camp and Fodder Supply**
 - **Health and Hygiene**
- **Way Forward**
 - » **Scientific mapping of Drought Prone areas**
 - » **Reducing the risk of droughts**:
 - Holistic and sustainable development **of watershed**;
 - **Improved water use efficiency in Agriculture**: Awareness, Micro-Irrigation; Changes in cropping pattern; rationalize electricity charges.
 - **Conservation of rainwater** in both urban and rural areas;
 - **Recycling of waste water**
 - **Properly implement building codes** to promote conservation and recycling.
 - **Afforestation activities** can increase water retaining capacity of the soil and also increases the chance of rainfall.
 - » **Developing a robust mechanism of EWS** in drought prone areas.
 - This will also require a robust methodology for drought declaration.
 - » **Awareness Generation**: Mission LIFE which promote sustainability among masses can go a long way in promoting water conservation and sustainable water utilization.
 - » **Capacity building** of institutions to respond to drought situation in an holistic manner.
 - » **More Research and Development**:
 - Effective Prediction of drought
 - Promoting less water use in washing machines, car washes etc.
 - Affordable, water conserving desalination plants.
 - » **Paris Agreement Targets**
 - The global efforts towards achieving "Net Zero" emission needs to be intensified to fight climate change in long run.
- **Conclusion**:

- » By adopting a holistic approach which includes hazard mapping, drought risk reduction efforts, developing a mechanism of EWS, people's participation, scientific advancements in desalination etc, India can reduce its vulnerability to the frequent occurrence of droughts in different parts of the country.

7. DAM SAFETY

- **Why in news?**
 - » In Oct 2023, Sikkim's highest Dam (Teesta-III Dam at Chungthang) was washed away after an GLOF which raised doubts about hydropower projects being developed in the country (Oct 2023)
 - Various reports have since revealed that there were no EWS, no risk assessment or preventive measures in place as required under the 2021 Dam Safety Act.
- **Example Questions**
 - » Discuss the key provisions of the Dam Safety Act 2021. How far does it go in ensuring structural and operational safety of dams [12.5 marks, 200 words]
 - » What are the key concerns related to Dam Safety in India? How far will the Dam Safety Act, 2021 be able to resolve these concerns? [10 marks, 150 words]
- **Introduction**
 - » Dams play a crucial role in the development of the country by supplying water for irrigation, flood control, and energy generation. India has 6,000 large dams in operation (which is third highest in the world). Further, around 4,00 large dams are under various phase of construction.
 - » However, a **poorly maintained and ill-operated dam** can become a source of threat not only for human life and infrastructure, but also for the environment. Therefore, there has been a long felt need of a **uniform law and administrative structure** in the country for the purpose of dam safety.
- **Key concerns associated with Dam Safety in India**
 - » **Very Old Dams:** around 4% (227) of large dams are more than 100 years old and 80% are more than 25 years old.
 - » **Earthquake prone areas:** Dams in Himalayas and other earthquake prone areas are under constant threat.
 - » **History of accidents:** India has faced **36 major dam failure in the past**, the worst one of Machchhu Dam (Gujarat) in 1979 in which about 2000 people had died.
 - » **Inadequacies** in maintaining Dam health and Safety:
 - **Poor Implementation** of the existing safety provisions under DAM Safety Act, 2021
 - A **report by CAG** has found that the structural strength of 348 large dams are suspect and they have not been inspected for over a decade.
 - Similarly, the **world bank report on Dam Rehabilitation and Improvement Project (DRIP)** indicates that the implementation of the program has been **moderately unsatisfactorily**.
- **Institutional Framework/Programs/Schemes dealing with dam safety in India**
 - » **The Central Water Commission**, Ministry of Jal Shakti through the National Committee on Dam Safety (NCDS), NDSO, SDSCO etc has been making constant endeavours in the direction of Dam Safety.

- » **Dam Rehabilitation and Improvement Project (DRIP)** is being implemented by Ministry of Jal Shakti with assistance from World Bank.
 - The main objectives of DRIP are:
 - a. TO improve the safety performance of selected existing dams (223 dams across 7 states) in a sustainable manner
 - b. To strengthen the dam safety institutional set up in participating states as well as at central level.
 - The seven DRIP states are - Uttarakhand, Madhya Pradesh, Jharkhand, Odisha, Karnataka, Kerala and TN.
- » **The Dam Health & Rehabilitation Monitoring Application (DHARMA)**
 - It is a webtool/app which is focused on digitizing dam related data effectively. It will help in easy identification of vulnerable dams and ensure need based rehabilitation.
- » **Ministry of Power and DRDO have signed an MoU for vulnerable Hydro Projects/ Power Stations** in Hilly Areas
 - Under this they would work jointly together towards developing suitable mitigation measures against avalanches, landslides, glaciers, glacial lakes, and other geo-hazards
 - For vulnerable projects in hilly areas, expertise of DRDO will be used for developing comprehensive EWS.

- **Dam Safety Act, 2021**

- » The act is aimed at helping states and UTs to adopt uniform safety procedure and thus ensure safety of the dams. It also gives **statutory backing** to various dam safety institutions and provides for strict punishment in case of the violation of the law.
- » It provides for surveillance, inspection, and maintenance of all specified dams across the country.
 - These dams are with height more than 15 meters, or height between 10 meters to 15 meters but with certain design and structural principle.
 - The act establishes a **robust Institutional Framework for Dam Safety**:
 - It sets up **two national bodies**
 - i. The **National Committee on DAM SAFETY** which would evolve policies and recommend regulations regarding dam safety.
 - ii. The **National Dam Safety Authority** which would implement policies of the National Committee, provide technical assistance to State Dam Safety Organizations (SDSO) and resolve matters (dispute resolution) between SDSOs of states or between SDSOs and Dam Owners.
 - The law also sets up **two state bodies**
 - i. **State Committee on Dam Safety** which will review work of SDSO, order Dam Safety Investigation, recommend dam safety measures and review the progress of such measures.
 - ii. **State Dam Safety Organization (SDSO)** will be responsible for surveillance, inspection, monitoring, operation, maintenance and investigation of dams.
- **Jurisdiction over dams**

- All specified dams will fall under jurisdiction of the SDSO of the state in which dam is situated.
- For dams owned by CPSU or which extends in two or more states or when a dam owned by one state is situated in other state, NDSA will have the jurisdiction and will play the role of SDSO.
- **What are states required to do?**
 - Provisions require states to classify dams based on hazard risk, conduct regular inspections, create emergency action plan, institute emergency flood warning systems, undertake safety reviews and period risk assessment studies.
- **Duties and Functions of DAM owners** (sufficient funds, trained manpower, dam safety units to conduct regular inspections, mandatory presence of engineers during floods and emergency, install emergency flood warning system; carry out risk assessment at regular intervals)
- **Comprehensive DAM Safety Evaluation (CSE)**
 - The act provides for comprehensive safety evaluation by independent panel of experts at regular intervals.
- **Offences and Penalties** for violation of provisions
- **Analysis of the act : Key challenges/Limitations**
 - » **Jurisdiction of Parliament on the issue (Entry 17 of the State List read with Entry 56 of the Union List,** gives powers to state to make laws on water supply, irrigation and canals, drainage and embankments, storage etc for intra state rivers).
 - » The **functions of the NCDS, NDSA, SCDS are listed in the schedule** of the act which can be modified by government through notification. Experts have raised concerns over this kind of overwhelming powers with central government.
 - » **States Raising Concerns** regarding NDSA having jurisdiction over dams owned by one state but situated in others. Some states feel that this takes away rights of states over their dams.
 - Note: TN own dams in the state of Kerala (in Mullaperiyar, Parambikulam, Tunakadavu, and Peruvaripallam)
 - » **States lack technical capability** to really implement the act in terms of number of trained personnel's, engineers etc.
 - The Sikkim GLOF reveals poor compliance at all levels of dam safety, from the dam's design to the spillway capacity.
 - » **Environment Impact ignored:** The act does not contain any norms which relates to environmental impact in the upstream and downstream of the rivers.
 - » **Lack of focus on operational safety** (like rate of filling or rate of water release) could lead to continuance of cases of Dam induced floods (e.g. Kerala floods, 2018)
- **Other problems related to Dam**
 - » **Lack of coordination between states** leads to faulty management of dams.
 - For e.g., the recent floods in Odisha was caused by faulty management of Hirakud Dam. One of the reasons for it was lack of information from Chhattisgarh to Odisha regarding the flow of water.

- **Way forward**
 - » **Set up the institutional framework envisaged under the law:** Dam Safety Policy should be finalized quickly to act as a guiding principle towards protection of Dams.
 - » **Promote More transparency:** Dam Safety is a public purpose and thus everything about dam safety, functions of institutions, their reports, decision minutes and agendas, everything should be promptly available to public.
 - » **Human Resource development:** We will need huge human resource for ensuring that trained people man dams, engineers are available for inspection and monitoring, emergency action plan etc.
 - » **Land use plans** should have dam safety issues integrated in it.
 - » **Operational Safety and Environment Impact** needs to be better integrated in the act and any future policies.
 - » **Increased coordination between states:**
 - E.g. of the **United States web-based integrated risk management tool** called **Dam Sector Analysis tool**. The tool was developed using variables from dam failure models and decision support systems, which enables the software to project downstream risk in the case of a dam failure.
- **Conclusion:**
 - » India's first prime minister, Pandit Jawaharlal Nehru, had referred to dams as the 'temples of modern India'. These temples would remain a boon, only if all the stakeholders work towards eliminating risks associated with Dam Safety.

8. FIRE SAFETY

- **Introduction: Context Based:**
 - » Two fire accidents in May 2024, one in Delhi's private nursing home killing 8 people and another in Game zone in Gujarat killing 33 people have highlighted the vulnerability of Indian cities to fire accidents.
- **Institutional Framework/Initiatives for fire safety in India:**
 - » **Fire Service is a state subject** in India and has been included as **municipal function in the 12th schedule** of the constitution. The municipal corporations and local bodies are responsible for providing fire services in many states.
 - » **Part 4 of the National Building Code, 2016** is titled 'Fire Safety and Life Safety' and covers requirements for fire prevention, life safety in relation to fire and fire protection of buildings.
 - » **Model Building Bylaws, 2003** also provides provisions related to fire safety, including the approval of building plan by Chief Fire Office.
 - » **"Scheme for Expansion and Modernization of Fire Services in the States"** - Launched by Ministry of Home Affairs in July 2023.
 - It has a total outlay of Rs 5,000 crores.
 - Objective of the scheme is to modernize Fire Services within National Disaster Response Fund (NDRF) with a view that activities for strengthening of fire services at the state-level through preparedness and capacity building component of the NDRF will be ensured.
- **Key reasons for Increasing fire accident cases:**
 - » **Poor implementation of fire safety norms** in buildings:
 - Fire safety is not given high priority in building plans.

- Poor compliance, specially in older buildings in the form of non-compliant structures, lack of precautionary maintenance like upkeep of extinguishers etc; gross overlook of safety procedures such as evacuation drills etc.
 - E.g. in May 2024 hospital fire in Delhi, emergency exits were absent, fire extinguishers were non functioning, lack of operational fire alarm.
 - » **Increasing Fire-Load of Indian Buildings:** An IIT Gandhinagar study has found that fire load in Indian cities today is 3 times of what was seen in late 1990s. It is due to use of false ceiling panel, electrical equipment, plastic door etc.
 - » **Electrical issues:** Faulty wiring, overloaded circuits, substandard wires and equipment etc are major initiators of fire accidents.
 - » **Negligence and human errors:** Improper handling of inflammables; unsafe practices like cooking near inflammables etc.
 - » **Unplanned Cities:** Hanging wires, storage of inflammable material, blocked pathways etc are common in various parts of the Indian cities making them highly vulnerable to fire accidents.
 - Small-crowded streets, unplanned colonies, narrow streets, incorrect parking, crowded markets etc. make it **difficult for fire-brigade trucks** difficult to reach the accident sites
 - » **Fire Safety audits lack legal backing:** Absence of clear provisions in any of the safety legislations regarding the scope, objective, methodology and periodicity of fire safety audits.
 - » **Deficiency of resources in Urban Local Bodies:** Thus ULBs in India are ill equipped in providing adequate fire safety cover to the population.
 - » **Climate Change** is also causing high temperatures and very dry weather. It increases the chance of fire accidents.
- **Way Forward**
- » **Comprehensive Risk-Assessment of our cities**
 - » **Fire Safety has to imbibed in the process** from the planning phase
 - Step to make fire safety a necessary part of B-Arch is a step in the right direction.
 - » **Strict implementation of the National Building Code, 2016**
 - NBC is a detailed set of guidelines for the construction, maintenance and operation of buildings of all kinds (residential, educational, institutional, assembly, mercantile, industrial, business, etc.) and includes a separate and comprehensive chapter on fire and life safety.
 - The National Building Code specifies, for instance, how many exits should be provided in a specific kind of building and where they must be placed.
 - » **Reducing the fire-load of Indian Buildings:** Building should be constructed with fire-resistant/retardant materials and there should be provisions for smoke detection systems and fire alarms.
 - » **Awareness Generation among masses:** Through firefighting workshops periodically in localities, schools, etc; Focus on how to handle inflammable material, why good quality wiring is crucial etc.
 - » **Empowering local bodies** with more resources and **modernization of fire departments** for quick response.
 - » **Legal backing to mandatory FSA:** It should provide for scope, methodology and periodicity
 - » **R&D** for developing for developing affordable, aesthetic and fireproof building material.

9. CHEMICAL DISASTER

- Introduction

- » Chemical disaster includes accidental release of toxic chemicals into the environment, resulting in death or injury to workers or members of nearby communities.
 - E.g.
 - Bhopal Gas Tragedy of 1984 - release of **methyl isocyanate** from the Chemical plant in Bhopal - killed at least 2,000 people.
 - Vizag gas leak of May 2020 - release of **styrene gas** from a Chemical Plant in Vizag - killed at least 12 people.
- » **Industrial accidents** include chemical disasters as well as other machine failures/blasts/mishappening in various industries. Gas leak, explosive blasts etc are included in it.

- **Institutional Mechanisms to deal with Chemical Disasters in India**

- i. **Ministry of Environment and Forest (MoEF)** is the nodal Ministry for the management of chemical disaster.
- ii. **NDMA** has prepared guidelines to direct various ministries, department and state authorities for the preparation of disaster management plans.

- **Factors that make India more Vulnerable to Chemical Disaster**

- i. **Non Availability of Accident Investigation Agency in India and Exclusive Chemical Accident Database**
 - They would have played an important role in learning lessons from past and investigate the root cause of the accidents and prevent its recurrence.
- ii. **Industrial Factors**
 - **Lack of modernization in Industries and poor maintenance of equipment**
 - e.g. In Vizag gas leak, the tank was an old one without adequate temperature monitoring apparatus
 - **Design Defects**
 - A lot of times even the new machines don't satisfy all the regulatory mechanisms and lead to chemical leaks and accidents
 - **Poor training of workforce** - Human errors cause industrial accidents.
 - **Non-compliance with SOPs**
 - This also results due to poor implementation of regulatory mechanisms.
 - **Lack of focus on out-of-factory disasters**
 - e.g. there is almost no mechanism to deal with leakage in case it happens during transportation.
 - **Poor Disposal Mechanism**
 - Hazardous waste processing and its proper disposal needs special attention as these activities can also contribute to fire, explosions and toxic releases to the environment.
- iii. **Regulatory Inefficiencies** - For instance, LG Polymers which accidentally released styrene gas, was also operating without requisite environment clearance.
- iv. **Terrorism and sabotage:** In the last two decades, threat of terrorism has also become a major worry in preventing chemical disasters.
- v. **Natural Disasters** which have also increased in frequency may also lead to chemical disasters

- e.g. Fukushima nuclear leak or the recent fire in chemical industry in Texas during Cyclone
- vi. **High Population density** especially around industrial hubs
- **Prevention and Response**
 - As far as chemical accidents are concerned, proactive approach and prevention is way better than the reactive approach. Each of the following stakeholders can play a role:
 - i. **Role of Industry**
 1. **Identification of Hazardous Activities**
 - A knowledgeable and dedicated team of qualified professionals to evaluate the hazards and risks arising from the day to day activities is essential.
 2. **Maintenance of Plant Facilities and Equipment**
 - Proper maintenance
 - Regular site safety and health inspection needs to be carried out to ensure that the plant facility is safe and all the equipment's are operating at the intended method.
 3. **Installation of vapor/gas detection system**
 4. **Compliance with existing rules and regulations**
 - Various rules and regulations of the state and centre government should be strictly adhered for a sustainable and safe process
 5. **Development of Human Resource Management**
 6. **Emergency Preparedness**
 - The industry must have a good emergency response team that can react swiftly to mitigate propagating disasters.
 - ii. **Role of Government**
 1. **Strictly enforcing all the safeguards**
 2. **Setting up of accident investigation Board and Chemical Accident Database**
 - An accident investigation board in similar lines of Chemical Safety Board of USA can be set up in India to investigate the chemical disaster and bring out guidelines based on the lessons learn in each accident would be helpful in preventing the recurrence.
 3. **Awareness Campaign**
 - Awareness regarding hazards arising out of a chemical disaster to the workers as well as public.
 - A good knowledge about the hazards by the workers themselves would help in reducing unsafe acts as well as in tackling the disaster if it were to happen.
 4. **Research and Development**
 - R&D towards newer technologies that can minimize the toxicity of the by-products of chemical industries can be carried out by the government.
 5. **Offsite Emergency Planning**
 6. **Dealing with transportation of Hazardous Chemicals**
 - Swift and timely availability of emergency response for disaster during transportation of hazardous chemical will help in mitigating and rescue the public involved.

iii. **Role of Public**

1. A general awareness of the risk associated with any chemical accident would help in reducing the outcome of the accident.
2. A mutual aid group can be set up to organize the general public in case of any disaster and provide training and awareness about the potential actions to be taken in case of chemical leakage.

iv. **Some other suggestions by Committee to investigate Vizag gas leak**

- **Environment** should be included in '**Concurrent List**' of the Constitution of India. This will empower states to enact laws according to the requirements (local conditions, pollution problems etc.)

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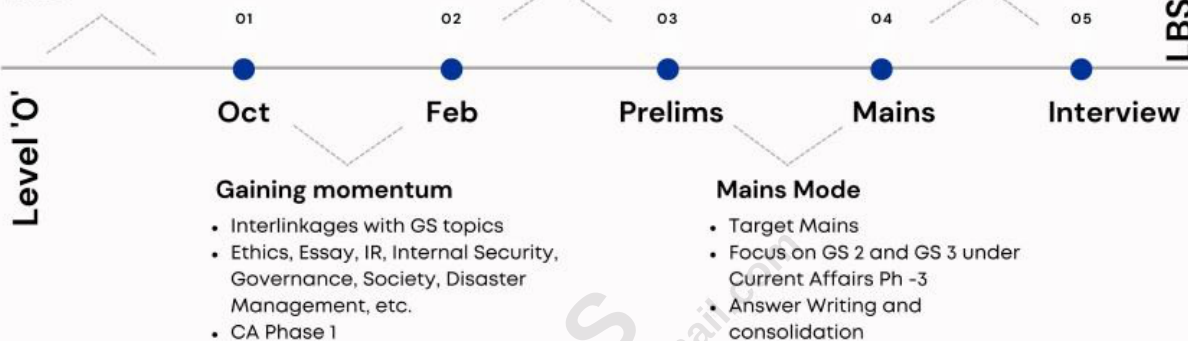
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